

Answer Key — Biology Grade 11

Sub-Strand 1.3: Taxonomy II • Lesson 4: Using Dichotomous Keys

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	D	C	A	A	C	B

USE AFTER: PREDICT PHASE — Predict: writing instructions so a farmer in Kericho gets the same name for a plant

Q1. A farmer in Kericho cannot ask you for help, but she must name a plant using only your written instructions. Which first step would most likely give her the same answer you get?

- A. Is the plant big or small?
- B. Is the plant useful as food?
- C. Is the plant a pretty colour?
- D. Does the leaf have parallel veins or net-like veins?**

Correct answer: D. Vein pattern is a visible feature that anyone can check and that does not change with opinion. Size, usefulness and prettiness depend on the person, so different people would answer differently.

USE AFTER: OBSERVE PHASE — Part A: reading the first couplet of the sample key and modelling it with moss

Q2. The first couplet of a key says: 1a: Plant has no true roots, stems or leaves. 1b: Plant has true roots, stems and leaves. Which plant belongs at 1a?

- A. Bean plant
- B. Fern
- C. Maize plant
- D. Moss**

Correct answer: D. Moss (Bryophyta) has no true roots, stems or leaves, so it fits 1a. Fern, bean and maize all have true roots, stems and leaves and fit 1b.

USE AFTER: OBSERVE PHASE — Part B: measuring features such as leaf length with rulers while using the key outside

Q3. A learner lays a ruler beside a sukuma wiki leaf. The base of the leaf is at the 2 cm mark and the tip is at the 14 cm mark. What is the length of the leaf?

- A. 14 cm
- B. 2 cm
- C. 12 cm**
- D. 16 cm

Correct answer: C. The leaf does not start at zero, so subtract the start reading from the end reading: 14 cm - 2 cm = 12 cm.

USE AFTER: EXPLAIN PHASE — Step 1: Think-Pair-Share on why the key gave the same answer to different people

Q4. Different groups used the same key on the same plants and got the same names. Their own earlier groupings by size, colour and usefulness did not agree. Why did the key give the same answer to everyone?

- A. It uses pairs of contrasting, observable features, so each person makes the same clear choice at each step.**
- B. It lets each person choose any number of statements that they like.
- C. It uses the colour of the plant, which is the same for all people.
- D. It groups plants by how useful they are, which everyone agrees on.

Correct answer: A. A key works for everyone because each couplet offers two contrasting statements about features that can be seen or measured. Colour, size and usefulness depend on opinion, so they gave different groupings.

USE AFTER: EXPLAIN PHASE — Step 2: listing what makes each couplet good and building the principles of construction

Q5. Which of these is the best couplet to include in a dichotomous key for plants?

A. 1a: Leaf has parallel veins. 1b: Leaf has net-like veins.

B. 1a: Plant is useful. 1b: Plant is not useful.

C. 1a: Plant is beautiful. 1b: Plant is not beautiful.

D. 1a: Leaf is green. 1b: Leaf is dark green.

Correct answer: A. A good couplet has two contrasting statements about observable features, and only one can be chosen. Parallel veins and net-like veins meet this rule. Usefulness and beauty are opinions, and green versus dark green do not clearly contrast.

USE AFTER: EXPLAIN PHASE — Step 3: Dichotomous-key Challenge, linking features to Bryophyta, Pteridophyta, monocots and dicots

Q6. A group is building a key for a maize plant, a grass plant, a bean plant and a sukuma wiki plant. They find that the leaves of maize and grass have parallel veins. To which group do maize and grass belong?

A. Dicots, because they are in Spermatophyta

B. Pteridophyta, because they have long leaves

C. Monocots, because parallel veins are a monocot feature

D. Bryophyta, because they are small green plants

Correct answer: C. Parallel veins are a feature of monocots such as maize and grass. Bean and sukuma wiki have net-like veins and are dicots. Bryophyta and Pteridophyta are different divisions with no seeds.

USE AFTER: MODEL BUILDING PHASE — Model revision: crossing out or relabelling groupings based on opinion, such as 'good to eat'

Q7. A group revises its classification model after using the key. Their old model has a group called 'plants that are good to eat'. What is the best revision?

A. Make the 'good to eat' group larger by adding more plants.

B. Cross out the group and replace it with observable features such as parallel or net-like veins.

C. Colour the group in red so that it stands out.

D. Keep the group, because food use is the most important feature of a plant.

Correct answer: B. Being good to eat is an opinion or use, not an observable feature, so different people group plants differently. Observable features give groups that anyone can check and that fit the key.